

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Find the stationary points and classify each using the second derivative.

[6 marks]

$$f(x) = x^3 - 3x^2 + 4$$

2

Use the chain rule to differentiate, then evaluate at $x=0$.

[5 marks]

$$y = \sin(2x^2 + x)$$

3

Prove by induction that the statement holds for all positive integers n .

[6 marks]

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}$$

4

A curve is defined implicitly. Find dy/dx in terms of x and y .

[5 marks]

$$x^2 + xy + y^2 = 7$$

5

Find the area enclosed between the two curves.

[6 marks]

$$y = x^2, \quad y = 2x$$

6

Evaluate the limit and justify any algebraic simplification.

[4 marks]

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

7

Differentiate the function and simplify your answer.

[5 marks]

$$y = (3x^2 + 1)^5$$

8

A rectangle has perimeter 40 cm. Find the dimensions that maximise its area, and show your result is a maximum.

[6 marks]
